

PROBABILITY 3: ASSIGNMENT 6

Problem 1. Let $A_1, A_2, \dots$ be a sequence of independent events, satisfying $P(A_n) < 1$ for all $n \geq 1$, and $P(\bigcup_{n=1}^{\infty} A_n) = 1$. Show that $P(\limsup_{n \rightarrow \infty} A_n) = 1$.

Problem 2. Give example of an infinite sequence of pairwise independent random variables for which Kolmogorov's zero-one law fails.

Problem 3. Suppose $(X_n)_{n \geq 1}$ is a sequence of random variables on a probability space such that $X_n \xrightarrow{P} X$. Show that there is a subsequence $(X_{n_k})_{k \geq 1}$ such that $X_{n_k} \xrightarrow{\text{a.s.}} X$.

Problem 4. Let $(\xi_n)_{n \geq 1}$ be a sequence of independent Bernoulli random variables with

$$\mathbb{P}(\xi_n = 0) = \mathbb{P}(\xi_n = 1) = \frac{1}{2}, \quad n \geq 1.$$

Given any $n \geq 1$, let Z_n denote the length of the maximal block inside the set of values $(\xi_1, \xi_2, \dots, \xi_n)$ that contains only 1's. Prove that

$$\frac{Z_n}{\log_2 n} \xrightarrow{P} 1.$$

Problem 5. Let $(X_n)_{n \geq 1}$ be independent $\mathcal{N}(0, 1)$ random variables, and let

$$M_n = \max\{X_1, \dots, X_n\}.$$

Show that

$$\frac{M_n}{\sqrt{2 \log n}} \xrightarrow{P} 1 \quad \text{as } n \rightarrow \infty.$$

Problem 6. Let ξ, ξ_n be i.i.d. random variables with $\mathbb{E}[\log_+ \xi] < \infty$ and $\mathbb{P}(\xi = 0) < 1$.

(1) Show that

$$\limsup_{n \rightarrow \infty} |\xi_n|^{1/n} = 1 \quad \text{a.s.}$$

(2) Let c_n be fixed real numbers. Show that the radius of convergence of the random power series $\sum_{n=0}^{\infty} c_n \xi_n z^n$ is almost surely equal to the radius of convergence of the non-random power series $\sum_{n=0}^{\infty} c_n z^n$.

Notation: For $x > 0$, $\log_+ x := \max\{x, 0\}$.

Problem 7. (Erdos-Renyi random graph) Let $V_n = \{1, 2, \dots, n\}$. Fix $0 < p \leq 1$. Let $X_{i,j}$, $1 \leq i < j \leq n$, be i.i.d. $\text{Ber}(p)$ random variables. Consider the random graph $G(n, p)$ whose vertex set is V_n and whose edges consist of pairs $\{i, j\}$ such that $X_{i,j} = 1$.

(1) Show that

$$\mathbb{P}\{G(n, p) \text{ is not connected}\} \leq \frac{1}{2} \sum_{k=1}^{n-1} \binom{n}{k} (1-p)^{k(n-k)}.$$

(2) Show that $\mathbb{P}\{G(n, p) \text{ is connected}\} \rightarrow 1$ as $n \rightarrow \infty$.

Problem 8. Let $X_1, X_2, \dots$ be iid random variables with mean μ and finite variance σ^2 . Write

$$S_n = \sum_{k=1}^n X_k.$$

(a) Compute $\mathbb{E}S_n$ and $\text{Var}(S_n)$.

(b) Prove that, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

(c) Prove that

$$\frac{S_{k^2}}{k^2} \longrightarrow \mu \quad \text{almost surely.}$$

(d) If $X_i \geq 0$ almost surely, deduce that

$$\frac{S_n}{n} \longrightarrow \mu \quad \text{almost surely.}$$

Problem 9. Let X_n be independent Bernoulli random variables (taking values 0 and 1) with

$$\mathbb{P}(X_n = 1) = n^{-\alpha}, \quad \alpha > 0.$$

(a) Show that $X_n \rightarrow 0$ in probability for every $\alpha > 0$.

(b) Determine for which α we have $X_n \rightarrow 0$ almost surely.

(c) When $0 < \alpha \leq 1$, prove that

$$\limsup_{n \rightarrow \infty} X_n = 1, \quad \liminf_{n \rightarrow \infty} X_n = 0 \quad \text{almost surely.}$$

(d) Which of your arguments require independence?

Problem 10. Let $X_1, X_2, \dots$ be independent random variables with

$$\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}, \quad S_n = \sum_{i=1}^n X_i.$$

(a) Show that

$$\mathbb{E}e^{tS_n} = (\cosh t)^n,$$

where $\cosh t = (e^t + e^{-t})/2$.

(b) Prove that $\cosh t \leq e^{t^2/2}$, prove that for $a > 0$,

$$\mathbb{P}(|S_n| \geq a) \leq 2e^{-a^2/(2n)}.$$

(c) Use this estimate and the Borel–Cantelli lemma to prove directly that $S_n/n \rightarrow 0$ almost surely.

(d) Show that for every fixed $C > \sqrt{2}$, almost surely

$$|S_n| < C\sqrt{n \log n}$$

for all sufficiently large n .